AN INFINITESIMAL CIRCULAR MORERA THEOREM

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Abstract. Morera's theorem characterizes holomorphicity of a continuous function through the vanishing of its integrals over a sufficiently rich family of closed curves. We establish a pointwise infinitesimal version in which only circles centered at each point and having arbitrarily small radius are tested. Let $D \subset \mathbb{C}$ be a domain and let $f \in C(D)$. We prove that if $\int_{|\zeta-a|=r} f(\zeta)\,d\zeta = o(r^2)$ as $r \to 0^+$ at every $a \in D$, then f is holomorphic in D. No uniformity in the center is assumed, and exact vanishing over all sufficiently small centered circles is included as a special case. The proof is entirely local. On each relatively compact subdomain, we first construct a continuous distributional primitive for the Wirtinger operator ∂. A polar-coordinate identity, extended from smooth functions to weak ∂-derivatives by mollification, converts the circular-integral hypothesis into an $o(r^2)$ deviation of the circular mean of the primitive from its value at the center. The Blaschke asymptotic mean-value criterion then makes the primitive's real and imaginary parts harmonic. Consequently the primitive is smooth, its ∂-derivative is holomorphic, and its distributional equality with f holds pointwise. This settles both the infinitesimal and exact alternatives in the local circular problem of Gaier and Zalcman.

1. Introduction

Throughout the paper, a domain is a connected open subset of the complex plane. We use

$$\partial = \frac{1}{2}(\partial_x - i\partial_y), \qquad \bar\partial = \frac{1}{2}(\partial_x + i\partial_y).$$

Morera's theorem is the classical converse to Cauchy's theorem: a continuous complex-valued function on a domain is holomorphic once its integrals over a sufficiently rich family of closed curves vanish. In one standard formulation it is enough to test boundaries of triangles contained in the domain; see, for example, Ahlfors [1]. Thus Morera's theorem converts an integral condition, imposed on closed curves, into the differential conclusion $\bar\partial f = 0$.

A natural refinement is to ask how much of the family of test curves is really needed. The problem considered here is a local circular version of this question. It was posed by D. Gaier and L. Zalcman and appeared as Problem 7.28 in Anderson–Barth–Brannan [2, Problem 7.28]; it is also recorded in Hayman–Lingham's problem collection [7, Problem 7.28]. In that formulation one asks whether a continuous function must be holomorphic if the integrals over all sufficiently small circles centered at the point in question vanish, and also whether the weaker infinitesimal condition of order $o(r^2)$ is already sufficient. The same problem points to Zalcman's work on analyticity, the Pompeiu problem, and mean-value equations [12, 13]. For neighboring recent work, Guo–He [6] resolve the area-measure interpretation of Hayman–Lingham Problem 7.29 by studying a variable-radius disk transform. That problem concerns area integrals over center-dependent disks and is geometrically distinct from the boundary-integral condition studied here, but both belong to the same group of Zalcman problems on integral rigidity. We use

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this circle of results only as context; the proof below is local and does not require the Fourier–Laplace machinery associated with the Pompeiu problem.

For $a \in \mathbb{C}$ and $r > 0$, let

$$B(a,r) = \{z \in \mathbb{C} : |z-a| < r\}, \qquad C(a,r) = \{z \in \mathbb{C} : |z-a| = r\},$$

where $C(a,r)$ is positively oriented. If u is continuous in a neighbourhood of $\overline{B(a,r)}$, set

$$A_u(a,r) = \frac{1}{2\pi}\int_0^{2\pi} u(a+re^{it})\,dt, \qquad J_u(a,r) = \int_{C(a,r)} u(\zeta)\,d\zeta. \tag{1}$$

If $D \subset \mathbb{C}$ is a domain, write $\rho_D(a) = \operatorname{dist}(a, \partial D)$, with the convention $\rho_D(a) = \infty$ when $D = \mathbb{C}$. All circular means and integrals below are taken only for radii $0 < r < \rho_D(a)$.

With the notation in (1), the scale r^2 is forced by the differentiable case. If $f \in C^1(D)$, Green's formula gives, whenever $\overline{B(a,r)} \subset D$,

$$J_f(a,r) = \int_{C(a,r)} f(\zeta)\,d\zeta = 2i\iint_{B(a,r)} \bar\partial f\; dA. \tag{2}$$

Consequently, (2) and the continuity of $\bar\partial f$ give

$$\frac{J_f(a,r)}{r^2} \longrightarrow 2\pi i\,\bar\partial f(a) \qquad (r \to 0^+). \tag{3}$$

Thus the hypothesis $J_f(a,r) = o(r^2)$ is exactly the infinitesimal assertion $\bar\partial f(a) = 0$ when the classical derivative exists. The point of the main theorem is that no differentiability assumption is needed.

This leads to the following infinitesimal circular version of Morera's theorem.

Theorem 1.1. *Let $D \subset \mathbb{C}$ be a domain and let $f \in C(D)$. Suppose that, for every $a \in D$,*

$$J_f(a,r) = o(r^2) \qquad (r \to 0^+,\ 0 < r < \rho_D(a)). \tag{4}$$

Then f is holomorphic in D.

As an immediate consequence, we obtain the corresponding exact local circular condition.

Corollary 1.2. *Let $D \subset \mathbb{C}$ be a domain and let $f \in C(D)$. Suppose that, for every $a \in D$, there is $\delta(a) > 0$ such that*

$$J_f(a,r) = 0 \qquad \text{whenever } 0 < r < \min\{\delta(a), \rho_D(a)\}.$$

Then f is holomorphic in D.

Proof. The stated vanishing implies (4) at every center. □

Theorem 1.1 and Corollary 1.2 give the asserted affirmative answer to the two alternatives in the Gaier–Zalcman problem.

1.1. **Proof strategy.** The proof does not differentiate f directly. On a relatively compact subdomain we first construct a continuous function F satisfying $\partial F = f$ in the sense of distributions. If F is smooth, a polar-coordinate calculation shows that the circular integral $J_f(a,s)$ is proportional to the derivative with respect to $\log s$ of the circular mean $A_F(a,s)$. The corresponding integrated identity remains valid under the weak hypothesis $\partial F = f$:

$$A_F(a,R) - F(a) = -\frac{i}{\pi}\int_0^R \frac{J_f(a,s)}{s}\,ds. \tag{5}$$

The assumption $J_f(a,s) = o(s^2)$ then implies $A_F(a,R) - F(a) = o(R^2)$ as $R \to 0^+$ at every point. A second-order asymptotic form of the mean-value characterization of harmonic functions shows that the real and imaginary parts of F are harmonic. Hence F is smooth,

∂F is holomorphic, and the distributional identity $\partial F = f$ becomes a pointwise identity by continuity.

1.2. **Organization of the paper.** Section 2 establishes the first two local tools used in the argument—a distributional ∂-primitive and the weak circular identity (5)—and recalls the asymptotic mean-value criterion for harmonicity. Section 3 proves Theorem 1.1. Section 4 records two comments on the scale r^2 and on the local nature of the hypothesis.

2. Preliminaries

This section contains only local facts. They are included in order to fix the normalizations of ∂, $\bar\partial$, and $d\zeta$, and to make the passage from weak derivatives to circular means explicit.

We use the standard distributional fundamental-solution identities

$$\bar\partial\left(\frac{1}{\pi z}\right) = \delta_0, \qquad \partial\left(\frac{1}{\pi \bar z}\right) = \delta_0. \tag{6}$$

For background on distributions and fundamental solutions one may consult Folland [5, Chs. 0–1] or Hörmander [8, Ch. I]. The first lemma is the corresponding local primitive construction for the operator ∂. Here $V \Subset \Omega$ means that $\overline{V}$ is a compact subset of Ω.

Lemma 2.1. *Let $\Omega \subset \mathbb{C}$ be open and let $h \in C(\Omega)$. If $V \Subset \Omega$, then there are an open set W and a function $U \in C(W)$ such that*

$$\overline{V} \subset W \Subset \Omega, \qquad \partial U = h \quad \text{in } \mathcal{D}'(W).$$

Proof. Choose an open set W with $\overline{V} \subset W \Subset \Omega$, and choose $\chi \in C_c^\infty(\Omega)$ such that $\chi = 1$ near $\overline{W}$. Extend χh by zero to $\mathbb{C}$, put $K(z) = 1/(\pi \bar z)$, and set $U = K * (\chi h)$. Since $K \in L^1_{\mathrm{loc}}(\mathbb{C})$ and χh is bounded and compactly supported, continuity of translations in $L^1_{\mathrm{loc}}(\mathbb{C})$ gives $U \in C(\mathbb{C})$. By (6),

$$\partial U = (\partial K) * (\chi h) = \delta_0 * (\chi h) = \chi h.$$

Since $\chi = 1$ on W, the identity $\partial U = h$ holds in $\mathcal{D}'(W)$. □

The next lemma is the analytic link between the circular integrals of a weak ∂-derivative and the circular means of its primitive. The proof is written in two stages: first the smooth calculation, where the sign and constant are visible, and then the standard mollification argument.

Lemma 2.2. *Let $\Omega \subset \mathbb{C}$ be open, let $U, h \in C(\Omega)$, and assume that $\partial U = h$ in $\mathcal{D}'(\Omega)$. If $a \in \Omega$, $R > 0$, and $\overline{B(a,R)} \subset \Omega$, then*

$$A_U(a,R) - U(a) = -\frac{i}{\pi}\int_0^R \frac{J_h(a,s)}{s}\, ds. \tag{7}$$

The integral on the right is absolutely convergent near 0.

Proof. We first note the endpoint estimate. Since $\int_{C(a,s)} d\zeta = 0$, we have $J_h(a,s) = \int_{C(a,s)} (h(\zeta) - h(a))\, d\zeta$. Put $\omega_h(a,s) = \sup_{|w-a|\le s} |h(w) - h(a)|$. Then, since h is continuous,

$$\left|\frac{J_h(a,s)}{s}\right| \le 2\pi \omega_h(a,s) \longrightarrow 0 \qquad (s \to 0^+). \tag{8}$$

Estimate (8) shows that the right-hand side of (7) is well defined at the lower endpoint.

Assume now that $U \in C^1(\Omega)$ and $h = \partial U$ classically. With $\zeta = a + se^{it}$, polar coordinates about a give

$$\partial = \frac{1}{2} e^{-it}\left(\partial_s - \frac{i}{s}\partial_t\right), \qquad d\zeta = ise^{it}\, dt.$$

Therefore, for $0 < s < R$,

$$\begin{aligned} J_h(a,s) &= \int_0^{2\pi} \partial U(a+se^{it})\, ise^{it}\, dt \\ &= \frac{is}{2}\int_0^{2\pi} \partial_s U(a+se^{it})\, dt + \frac{1}{2}\int_0^{2\pi} \partial_t U(a+se^{it})\, dt \\ &= i\pi s \frac{d}{ds} A_U(a,s), \end{aligned}$$

where the ∂_t-term integrates to zero over one period. Since $A_U(a,s) \to U(a)$ as $s \to 0^+$, integration from 0 to R yields (7).

It remains to pass to the stated weak case. Choose $R_1 > R$ such that $\overline{B(a,R_1)} \subset \Omega$, and choose $\chi \in C_c^\infty(\Omega)$ such that $\chi = 1$ in a neighbourhood of $\overline{B(a,R_1)}$. Extend χU and χh by zero to $\mathbb{C}$. Let φ_ε be a standard mollifier supported in $B(0,\varepsilon)$, and for $0 < \varepsilon < R_1 - R$ put

$$U_\varepsilon = \varphi_\varepsilon * (\chi U), \qquad h_\varepsilon = \varphi_\varepsilon * (\chi h).$$

For points of $B(a, R_1 - \varepsilon)$, all contributions to the convolution arise from the region where $\chi = 1$. Consequently $\partial U_\varepsilon = h_\varepsilon$ in $B(a, R_1 - \varepsilon)$. Since $\overline{B(a,R)} \subset B(a, R_1 - \varepsilon)$, this identity holds in a neighbourhood of $\overline{B(a,R)}$, and the smooth identity (7) applies to $(U_\varepsilon, h_\varepsilon)$ at radius R.

As $\varepsilon \to 0$, the functions U_ε and h_ε converge uniformly to U and h, respectively, on $\overline{B(a,R)}$. Moreover, for $0 < s \le R$,

$$\left| \frac{J_{h_\varepsilon}(a,s) - J_h(a,s)}{s} \right| \le 2\pi \|h_\varepsilon - h\|_{L^\infty(\overline{B(a,R)})}.$$

Hence

$$\int_0^R \left| \frac{J_{h_\varepsilon}(a,s) - J_h(a,s)}{s} \right| ds \le 2\pi R \|h_\varepsilon - h\|_{L^\infty(\overline{B(a,R)})} \to 0.$$

The right-hand sides of the mollified identities therefore converge to the right-hand side of (7); the left-hand sides converge by the uniform convergence of U_ε. This proves the identity. □

The last preliminary result is the classical Blaschke asymptotic mean-value criterion. The planar form below is precisely Kuznetsov [9, Thm. 1.6]. For related mean-value and generalized-Laplacian results, see Axler–Bourdon–Ramey [3, Thms. 1.4 and 1.24], Netuka–Veselý [10], Pokrovskii [11], and Córdoba–Ocáriz [4].

Lemma 2.3. *Let $\Omega \subset \mathbb{C}$ be open and let $u \in C(\Omega)$ be real-valued. If, for every $a \in \Omega$,*

$$A_u(a,r) - u(a) = o(r^2) \qquad (r \to 0^+), \tag{9}$$

then u is harmonic in Ω.

3. Proof of the main theorem

Let $V \Subset D$ be an arbitrary open set. By Lemma 2.1, applied with $h = f$, there are an open set W with $\overline{V} \subset W \Subset D$ and a function $F \in C(W)$ such that

$$\partial F = f \quad \text{in } \mathcal{D}'(W). \tag{10}$$

Fix $a \in V$. For all sufficiently small $r > 0$, $\overline{B(a,r)} \subset V$. Lemma 2.2, used with $U = F$ and $h = f$, gives

$$A_F(a,r) - F(a) = -\frac{i}{\pi}\int_0^r \frac{J_f(a,s)}{s}\, ds. \tag{11}$$

The hypothesis (4) is pointwise in the center. Hence, for this fixed a and for every $\eta > 0$, there is $\delta > 0$ such that $|J_f(a,s)| \le \eta s^2$ whenever $0 < s < \delta$. If $0 < r < \delta$, then (11) yields

$$|A_F(a,r) - F(a)| \le \frac{1}{\pi}\int_0^r \eta s\, ds = \frac{\eta}{2\pi} r^2.$$

Since η is arbitrary,

$$A_F(a,r) - F(a) = o(r^2) \qquad (r \to 0^+) \tag{12}$$

at the point a. As $a \in V$ was arbitrary, (12) holds at every point of V.

By (12), the hypothesis (9) of Lemma 2.3 holds for both $\operatorname{Re} F$ and $\operatorname{Im} F$. Hence F is harmonic componentwise in V. Therefore $F \in C^\infty(V)$, and

$$\bar\partial(\partial F) = \frac{1}{4}\Delta F = 0 \tag{13}$$

classically in V. By (13), ∂F is holomorphic in V.

It remains only to identify ∂F with the original continuous function f. By (10), the two functions agree as distributions on V. Hence they agree almost everywhere. Since f is continuous and ∂F is smooth, the difference $f - \partial F$ is a continuous function that vanishes almost everywhere; it must vanish everywhere. Thus $f = \partial F$ pointwise on V, and f is holomorphic on V. Because $V \Subset D$ was arbitrary, f is holomorphic in D. This proves Theorem 1.1.

4. Final remarks

We first make explicit the coefficient detected by the normalized circular integral in the differentiable case.

Remark 4.1. Formula (3) shows that, for C^1 functions, the quantity $r^{-2}J_f(a,r)$ detects the value of $2\pi i\,\bar\partial f(a)$. For example, if $f(z) = \bar z$, then $\int_{|\zeta-a|=r} \bar\zeta\, d\zeta = 2\pi i\, r^2$. Thus the order r^2 in Theorem 1.1 is the first nontrivial order at which non-holomorphicity can be seen in the smooth case.

We close by emphasizing the entirely local nature of the argument.

Remark 4.2. The proof uses only the little-oh condition at each fixed center a. No uniformity in a is assumed. The construction of the ∂-primitive is also local: it is made on relatively compact subdomains, and no global topological condition on D is involved.

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